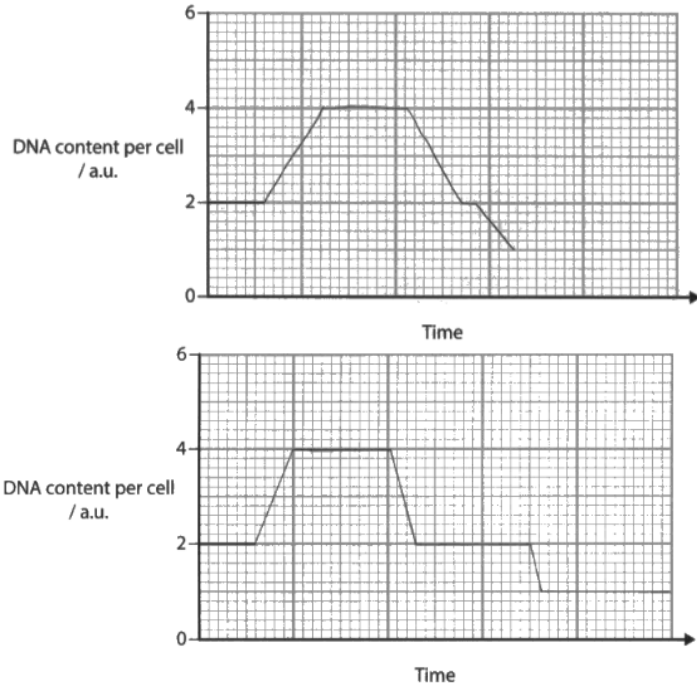


Question Number	Answer	Additional guidance	Mark
5(a)	• initial doubling to 4 a.u. (1) • staying at {4 a.u. / new DNA content} for a period of time and then {halving / returning back to 2 a.u.} for a period of time (1) • reduction of DNA content to {haploid / 1 a.u.} with no further {increase / decrease} (1)	 The top graph shows a curve that starts at 2, rises to 4, stays at 4, then falls back to 2 and continues to decrease. The bottom graph shows a curve that starts at 2, rises to 4, stays at 4, then falls back to 2 and stays at 2 for a short period before falling to 1.	(3)

Question Number	Answer	Additional guidance	Mark
5(b)	A calculation in which: • correct measurement and conversion (1) • correct magnification given to two significant figures (1)	<u>Example of calculation:</u> $16 \times 1000 = 16\,000\ \mu\text{m}$ ACCEPT tolerance of $\pm 0.5\ \text{mm}$ $16\,000 \div 24 = \times 670$ max 1 if given incorrect unit ACCEPT 2 sig fig answers in range 650 to 690 ACCEPT answers in standard form Correct answer scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
5(c)	A description that makes reference to five of the following points: • tube created due to release of {digestive / hydrolytic} enzymes (1) • {nucleus / nuclei / gamete(s)} transported to {ovule / ovary / micropyle / egg cell / polar nuclei / female gamete} (1) • generative nucleus divides to form two {male / haploid} nuclei (1) • double fertilisation occurs (1) • one (male) nucleus fuses with {egg cell / haploid / female} nucleus resulting in formation of zygote (1) • one male nucleus fuses with the two polar {nuclei / bodies} resulting in formation of endosperm (nucleus) (1)	REJECT pollen transported IGNORE gametes / sperm / cells REJECT forms two × {generative nucleus} REJECT diploid / meiosis REJECT generative nucleus {fusing / fertilizing} ecf for gametes / sperm IGNORE cells REJECT generative nucleus {fusing / fertilizing} ecf for gametes / sperm IGNORE cells	(5)

Question Number	Answer	Mark
5(d)(i)	The only correct answer is A 15 <i>B is not correct because $0.036 \times 418 = 15$ heterozygotes</i> <i>C is not correct because $0.036 \times 418 = 15$ heterozygotes</i> <i>D is not correct because $0.036 \times 418 = 15$ heterozygotes</i>	(1)

Question Number	Answer	Mark
5(d)(ii)	The only correct answer is C <i>A is not correct because leaf width shows continuous variation</i> <i>B is not correct because leaf width is determined by multiple alleles for several genes</i> <i>D is not correct because leaf width shows continuous variation</i>	(1)